Ex.No.1	**Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.**
Date:	

AIM: - To Study of various Network commands used in Linux and Windows

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	`arp -a`	`arp -a`	Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network.	`arp -a`	`Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic`
2	`hostname`	`hostname`	Displays the hostname (computer name) of the local system.	`hostname`	`DESKTOP-PC01`
3	`ipconfig /all`	`ipconfig /all`	Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address.	`ipconfig /all`	`IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes`
4	`nbtstat -a`	`nbtstat -a <hostname>`	Displays NetBIOS name table of a remote computer for troubleshooting NetBIOS over TCP/IP.	`nbtstat -a DESKTOP-PC01`	`NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered`
5	`netstat`	`netstat`	Displays active TCP/IP connections, listening ports, routing tables, and interface statistics.	`netstat`	`Proto Local Address Foreign Address State TCP 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
6	`netstat -r`	`netstat -r`	Displays the IP routing table of the local computer.	`netstat -r`	`IPv4 Route Table Network Destination Netmask Gateway Interface`
7	`nslookup`	`nslookup <domain>`	Queries the DNS server to obtain the IP address or DNS records of a domain.	`nslookup www.google.com`	`Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68`
8	`pathping`	`pathping <hostname/IP>`	Combines the features of `ping` and `tracert` to trace the route and measure packet loss at each hop.	`pathping www.google.com`	`Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%`
9	`ping`	`ping <hostname/IP>`	Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages.	`ping www.facebook.com`	`Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54`
10	`route`	`route print`	Displays or modifies the IP routing table. It is commonly used to view or configure static routes.	`route print`	`IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface`

Some important Linux networking commands

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	ip	`ip <OPTIONS> <OBJECT> <COMMAND>`	Displays and manages IP addresses, routing tables, and	`ip address show`	`2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			network interfaces.		
2	ip address add	ip address add <IP>/<Prefix> dev <Interface>	Assigns an IP address to a network interface.	ip address add 192.168.1.254/24 dev eth0	Assigned 192.168.1.254/24 to eth0 successfully.
3	ip address del	ip address del <IP>/<Prefix> dev <Interface>	Removes an IP address from a network interface.	ip address del 192.168.1.254/24 dev eth0	Deleted 192.168.1.254/24 from eth0.
4	ip link set up	ip link set <Interface> up	Brings the network interface online.	ip link set eth0 up	Interface eth0 is UP.
5	ip link set down	ip link set <Interface> down	Brings the network interface offline.	ip link set eth0 down	Interface eth0 is DOWN.
6	ip link set promisc on	ip link set <Interface> promisc on	Enables promiscuous mode on the interface.	ip link set eth0 promisc on	Promiscuous mode enabled on eth0.
7	ip route add default	ip route add default via <Gateway> dev <Interface>	Adds the default gateway for all outgoing traffic.	ip route add default via 192.168.1.254 dev eth0	Default route added successfully.
8	ip route add	ip route add <Network> via <Gateway>	Adds a static route through a gateway.	ip route add 192.168.1.0/24 via 192.168.1.254	Route added successfully.
9	ip route add (Direct Route)	ip route add <Network> dev <Interface>	Adds a direct route through a network interface.	ip route add 192.168.1.0/24 dev eth0	Route added via eth0.
10	ip route delete	ip route delete <Network> via <Gateway>	Deletes a static route.	ip route delete 192.168.1.0/24 via 192.168.1.254	Route deleted successfully.
11	ip route get	ip route get <Destination IP>	Displays the route used to reach a destination.	ip route get 10.10.1.4	10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100
12	ifconfig	ifconfig	Displays and configures network	ifconfig	eth0: flags=4163<UP,BROADCAST,RUNNING>

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			interface parameters (deprecated in modern Linux).		inet 192.168.1.100 netmask 255.255.255.0
13	mtr	mtr <hostname/IP>	Combines Ping and Traceroute to analyze network paths.	mtr google.com	HOST: localhost Loss% Snt Last Avg Best Wrst StDev
14	mtr -g	mtr -g <hostname>	Displays numeric IP addresses instead of hostnames.	mtr -g google.com	192.168.1.1 → 142.250.190.78
15	mtr -b	mtr -b <hostname>	Displays both hostnames and IP addresses.	mtr -b google.com	router.local (192.168.1.1) google.com (142.250.190.78)
16	mtr -c	mtr -c <count> <hostname>	Sends a specified number of probes.	mtr -c 10 google.com	10 probes completed successfully.
17	tcpdump -D	tcpdump -D	Lists all available network interfaces.	tcpdump -D	1.eth0 2.lo 3.wlan0
18	tcpdump	tcpdump -i <Interface>	Captures packets on a network interface.	tcpdump -i eth0	Listening on eth0...
19	tcpdump -c	tcpdump -i <Interface> -c <count>	Captures a specified number of packets.	tcpdump -i eth0 -c 10	10 packets captured.
20	tcpdump host	tcpdump -i <Interface> host <IP>	Captures traffic to and from a specific host.	tcpdump -i eth0 host 8.8.8.8	Captured packets to/from 8.8.8.8.
21	tcpdump src host	tcpdump -i <Interface> src host <IP>	Captures packets originating from a specific host.	tcpdump -i eth0 src host 8.8.8.8	Source packets from 8.8.8.8 displayed.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
22	tcpdump dst host	`tcpdump -i <Interface> dst host <IP>`	Captures packets destined for a specific host.	`tcpdump -i eth0 dst host 8.8.8.8`	`Destination packets to 8.8.8.8 displayed.`
23	tcpdump net	`tcpdump -i <Interface> net <Network>/<Prefix>`	Captures traffic from a specified network.	`tcpdump -i eth0 net 10.1.0.0/24`	`Packets from network 10.1.0.0/24 displayed.`
24	tcpdump port	`tcpdump -i <Interface> port <Port>`	Captures traffic on a specific port.	`tcpdump -i eth0 port 53`	`DNS packets captured.`
25	tcpdump host and port	`tcpdump -i <Interface> host <IP> and port <Port>`	Captures traffic for a specific host and port.	`tcpdump -i eth0 host 8.8.8.8 and port 53`	`DNS traffic with host 8.8.8.8 displayed.`
26	tcpdump port not	`tcpdump -i <Interface> port not <Port1> and not <Port2>`	Captures all traffic except the specified ports.	`tcpdump -i eth0 port not 53 and not 25`	`Traffic excluding ports 53 and 25 captured.`
27	ping	`ping <hostname/IP>`	Tests network connectivity using ICMP Echo Requests and Echo Replies.	`ping google.com`	`64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms`
28	ping -c	`ping -c <count> <hostname>`	Sends a specified number of ICMP Echo Requests.	`ping -c 10 google.com`	`10 packets transmitted, 10 received, 0% packet loss.`

Configuring an Ethernet connection by using nmcli:

Step No.	Task	Command	Expected Result
1	List all Network Manager connection profiles	`nmcli connection show`	Displays all available network connection profiles with their

Step No.	Task	Command	Expected Result
			Name, UUID, Type, and Device.
2	Create a new Ethernet connection profile It is specifically an **`nmcli` connection management command**. • **`nmcli`** → NetworkManager Command Line Interface • **`connection`** → Manages network connection profiles. • **`add`** → Creates a new connection profile.	`nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet`	Creates a new Ethernet connection profile named **Internal-LAN**.
3	Rename an existing connection profile (Optional)	`nmcli connection modify "Wired connection 1" connection.id Internal-LAN`	Renames the existing connection profile to **Internal-LAN**.
4	Display the current settings of the connection profile	`nmcli connection show "Internal-LAN"`	Displays interface name, autoconnect status, IPv4 and IPv6 configuration details.
5	Configure IPv4 using DHCP	`nmcli connection modify "Internal-LAN" ipv4.method auto`	Configures the interface to obtain the IPv4 address automatically using DHCP.
6	Configure a static IPv4 address	`nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com`	Assigns a static IPv4 address, gateway, DNS server, and DNS search domain.
7	Configure IPv6 using SLAAC	`nmcli connection modify "Internal-LAN" ipv6.method auto`	Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC).
8	Configure a static IPv6 address	`nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64`	Assigns a static IPv6 address, gateway, DNS

Step No.	Task	Command	Expected Result

		`ipv6.gateway 2001:db8:1::ffee`
		`ipv6.dns 2001:db8:1::ffbb`
		`ipv6.dns-search example.com`
			server, and search domain.

9	Activate the connection profile	`nmcli connection up Internal-LAN`	Activates the **Internal-LAN** network connection profile.
10	Verify the IP address configuration	`ip address show enp1s0`	Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface.
11	Verify the IPv4 default gateway	`ip route show default`	Displays the configured IPv4 default gateway.
12	Verify the IPv6 default gateway	`ip -6 route show default`	Displays the configured IPv6 default gateway.
13	Verify the DNS configuration	`cat /etc/resolv.conf`	Displays the configured DNS servers and DNS search domain.
14	Test network connectivity	`ping <host-name-or-IP-address>`	Sends ICMP Echo Requests to verify connectivity with the specified host.

Sample Verification Outputs

Verification Command	Sample Output
ip address show enp1s0	inet 192.0.2.1/24 inet6 2001:db8:1::fffe/64
ip route show default	default via 192.0.2.254 dev enp1s0 proto static
ip -6 route show default	default via 2001:db8:1::ffee dev enp1s0 proto static
cat /etc/resolv.conf	search example.com nameserver 192.0.2.200 nameserver 2001:db8:1::ffbb
ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms

RESULT: